Chain-of-Thought Protection Against J-Space Ablation Diminishes with Task Difficulty

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1 Hypothesis

My hypothesis is that chain of thought make models more robust to J-space ablation on easier problems, and this robustness should be diminishing as problem difficulty increases. The intuition for this is that chain of thought writes intermediate states into the transcript, reducing the need for the model to preserve them internally. This external record should be most useful when the required intermediate state is short and easy to express. As problems become harder and require larger, more complex intermediate states, the transcript is likely to become a less complete substitute for internal representations.

2 Experiment Design

I evaluate the robustness of J-Space with direct answers and answer generated with chain-of-thought under four ablation strengths: none, light over 3 layers, medium over 6 layers, and heavy over 12 layers.

As in the original paper, the unablated model's top 10 next token candidates are protected during ablation, preventing the intervention from trivially removing the token the model is about to generate.

To evaluate this hypothesis, I sample 90 questions from GSM8K, 90 from MATH-500, and 9 from AIME 2024 (the samples were kept low due to compute constraints). I assume the three datasets are ordered by difficulty, GSM8K $<$ MATH-500 $<$ AIME 2024. To obtain further fine grained effect, I further partition each dataset into 3 difficulty levels D1, D2 and D3 using the length of the gold solution as a proxy for problem difficulty.

I report accuracy across datasets, difficulty levels, and ablation strengths, together with Wilson 95% confidence intervals, paired lost and gained items, two sided binomial tests and some completion lengths (this causes token limit failures).

Since the prediction was that chain-of-thought prompting would exhibit high retention on GSM8K, with retention decreasing on the more challenging datasets, similar retention across datasets or comparable degradation for direct-answer and chain-of-thought prompting, would nullify the hypothesis.

3 Experimental Details

The experimentation was performend on Qwen3-4B in bfloat16 precision on a single H100 GPU using greedy decoding. The original study [1] does not specify the prompting setup or inference configuration used for the ablation experiments. I therefore disable the model's native reasoning mode, as otherwise the model could externalize its reasoning before producing the final answer, confounding the ablations.

Despite disabling reasoning mode and explicitly instructing the model to answer directly, Qwen3 4B occasionally produces brief step by step reasoning, even on relatively simple GSM8K problems. This behavior likely due the model's extensive post training and can we considered a limitation of the study.

I use the published `neuronpedia/jacobian-lens` artifact [2] for Qwen3-4B, which was fitted by its authors on 479 WikiText prompts across layers 0 to 34. I do not fit the J Lens locally. Following the original methodology, I measure the intervention band on 20 passages from WikiText 103.

The light ablation intervenes on layers 20 to 22, the medium ablation on layers 23 to 28, and the heavy ablation on layers 20 to 31. At each selected layer and token position, I remove the joint span of the 5 highest scoring unprotected J Lens directions after QR orthonormalization. During ablated decoding, I shadow an unablated forward pass to identify the protected next token candidates.

The GSM8K evaluation used 90 examples from the test split, with maximum generation lengths of 513 tokens for direct prompting and 1,023 tokens for chain of thought prompting. The MATH-500 evaluation used 90 test examples with a maximum generation length of 4,096 tokens. The AIME 2024 evaluation used 9 of the 30 problems from the combined AIME I and II sets, with a maximum generation length of 8,192 tokens. I used a dataset specific answer extractor for each benchmark.

4 Experimental Results

Dataset	Prompt	None	Light	Medium	Heavy	Retention
GSM8K	Direct	0.656	0.511	0.589	0.467	0.712
	Chain of thought	0.933	0.956	0.878	0.889	**0.952**
MATH-500	Direct	0.556	0.422	0.456	0.378	0.680
	Chain of thought	0.756	0.611	0.589	0.344	**0.456**
AIME 2024	Direct	0.222	0.111	0.111	0.222	1.000
	Chain of thought	0.111	0.222	0.222	0.222	2.000

Table 1: Exact-match accuracy. Retention is heavy accuracy divided by unablated accuracy.

Under all three ablation settings Chain-of-thought accuray retention is substantially higher on GSM8K compared to the more challenging MATH-500, dropping from 95.2% to 45.6%. In contrast, direct-answer retention remains stable across the two datasets, decreasing only slightly from 71.2% to 68.0%.

On GSM8K, chain-of-thought accuracy decreases from 93.3% [86.2%, 96.9%] to 88.9% [80.7%, 93.9%]. Six previously correct problems become incorrect after ablation, while two previously incorrect problems become correct, yielding a non-significant change ($p = 0.289$). On MATH-500, the effect is much larger: chain-of-thought accuracy falls from 75.6% [65.8%, 83.3%] to 34.4% [25.4%, 44.7%]. Thirty-nine previously correct problems become incorrect, while only two previously incorrect problems become correct, giving a highly significant change ($p = 7.8 \times 10^{-10}$).

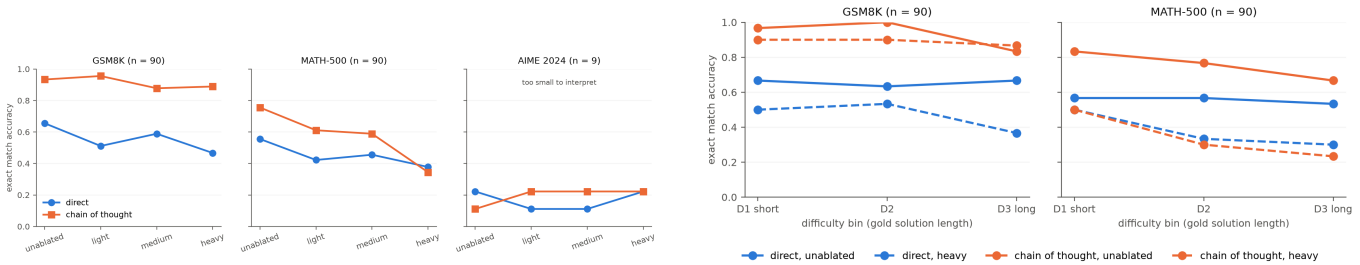
Dataset	Prompt	Unablated [95% CI]	Heavy [95% CI]	Lost / gained	p	Ret. (untrunc.)
GSM8K	Direct	0.656 [0.553, 0.746]	0.467 [0.367, 0.569]	21 / 4	0.00091	0.719 (n=88)
	Chain of thought	0.933 [0.862, 0.969]	0.889 [0.807, 0.939]	6 / 2	0.289	0.975 (n=87)
MATH-500	Direct	0.556 [0.453, 0.654]	0.378 [0.285, 0.481]	19 / 3	0.00086	0.680 (n=86)
	Chain of thought	0.756 [0.658, 0.833]	0.344 [0.254, 0.447]	39 / 2	7.8×10^{-10}	0.484 (n=81)

Table 2: Heavy ablation vs No ablation. Intervals are Wilson 95%; p is a two-sided binomial test over the items whose correctness changed, the last column rescores each pair after excluding items truncated at the token limit on either side. AIME 2024 is omitted due to reasons mentioned before.

During heavy ablation I observe that the mean completion length increases from 385 to 502 tokens on MATH-500, compared with only 136 to 146 tokens on GSM8K. This causes 8 of 90 MATH-500 chain-of-thought completions to reach the token limit. Excluding problems truncated in either paired condition still yields a retention rate of 0.484 over the remaining 81 items.

Within MATH-500, heavily ablated chain-of-thought accuracy decreases from 50.0% on easy problems to 23.3% on difficult problems. By comparison, unablated accuracy declines only from 83.3% to 66.7%. In contrast, the heavily ablated curve on GSM8K remains close to 90% across all solution-length bins.

AIME 2024 is excluded from interpretation because the baseline performance is only 2 correct answers out of 9 problems. A single problem alters the reported retention, making it unstable to support meaningful conclusions



(a) Accuracy across ablation strengths.

(b) Accuracy by gold-solution-length bin. Solid lines are unablated; dashed lines are heavily ablated.

Figure 1: Accuracy under J-space ablation. The nine-item AIME sample in panel (a) is shown for completeness only.

5 Analysis of Results

The between-dataset comparison cleary supports the hypothesis. Under the heavy ablation, chain of thought retention significantly drops on MATH500 as compared GSM8K indicating that chain of thought induced robustness in J-Space ablation is problem difficulty depended.

One plausible reason for this seems to be that protection depends on the form of the intermediate state. Simple numerical

state may be easy to preserve in text, while symbolic or structural state may still depend on internal representations.

The within-dataset evidence is more limited. The expected decline appears on MATH-500 but not on GSM8K. I think this is because solution length is not a reliable proxy for difficulty. Kordi et al. use Item Response Theory to estimate difficulty from model response patterns rather than surface features such as solution length [3]. Due to compute constraints, I was not able to apply this approach.

There are several limitation to this study. First, as mentioned earlier use to heavy posttraining some of the direct answer have a chain of though like behaviour. The dataset set size is small. Finally, AIME 2024 which is the hardest dataset was dropped.

With more compute, I would run the experiment on a stronger model (32-72B), estimate inter dataset difficulty using Rasch model and use complete dataset. I would also include some non-math dataset that requires reasoning/thinking like ARC-AGI etc to see how the results extend.

References

[1] W. Gurnee, N. Sofroniew, A. Pearce, M. Piotrowski, I. Kauvar, R. Chen, A. Soligo, P. Bogdan, E. Ong, R. Wang, T. B. Thompson, D. Abrahams, S. Kantamneni, E. Ameisen, J. Batson, and J. Lindsey. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*, 2026. Available online.

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[3] Y. Kordi, N. V. Nayak, M. Zuo, I. Nguyen, and S. Bach. Revisiting generalization across difficulty levels: It's not so easy. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics*, pages 7014–7042, 2026. doi:10.18653/v1/2026.eacl-long.330.